SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: 1 **Total Marks:**

Code of Conduct:

I POOJASREE B (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: POOJASREE B (192411091)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Case Study: Web Development Standards Implementation

Introduction

A startup is developing a modern web application and faces several issues related to HTML5 elements, HTTP methods, semantic HTML, XML syntax, and web communication protocols. These technologies are fundamental to modern web development because they determine how web content is embedded, how data is transferred, how webpages are structured, how configuration files are written, and how browsers communicate with web servers.

The correct implementation of these standards is important for creating a web application that is functional, accessible, secure, maintainable, and compatible with different browsers and systems.

1. Select the correct HTML tag for video embedding and justify your answer.

Correct Answer: B. `<video src="video.mp4" controls></video>`

The correct HTML5 element for embedding a video in a webpage is the `<video>` element.

```
<video src="video.mp4" controls></video>
```

The `<video>` element was introduced in HTML5 to allow developers to embed video content directly into webpages without depending on external plugins.

The `src` attribute specifies the location of the video file:

```
src="video.mp4"
```

The `controls` attribute displays built-in video controls such as:

- Play
- Pause
- Volume control
- Progress bar
- Full-screen option

Therefore, the user can interact directly with the video through the browser.

A more complete implementation can be:

```
<video controls width="600">
    <source src="video.mp4" type="video/mp4">
    Your browser does not support the video element.
</video>
```

Using the `<source>` element allows multiple video formats to be provided:

```
<video controls>
    <source src="video.mp4" type="video/mp4">
    <source src="video.webm" type="video/webm">
</video>
```

The browser can select a supported format.

Analysis of the other options

Option A: `<media>`

```
<media src="video.mp4"></media>
```

`<media>` is not a standard HTML5 element for embedding videos.

Option C: `<movie>`

```
<movie src="video.mp4"></movie>
```

`<movie>` is not a valid standard HTML5 element.

Option D: `<vid>`

```
<vid src="video.mp4"></vid>
```

`<vid>` is also not a standard HTML element.

Conclusion

The `<video>` element is the correct choice because it is a standard HTML5 element specifically designed for embedding video content. It supports native browser controls, improves compatibility, and eliminates the need for additional plugins.

2. Identify the appropriate HTTP method for form submission and explain why.

Correct Answer: B. POST

For submitting registration data, the appropriate HTTP method is **POST**.

Example:

```
<form action="/register" method="POST">
    <input type="text" name="name">
    <input type="email" name="email">
    <button type="submit">Register</button>
</form>
```

When the form is submitted, the browser sends the data to the server using an HTTP POST request.

A simplified request may look like:

```
POST /register HTTP/1.1
Host: example.com
Content-Type: application/x-www-form-urlencoded

name=John&email=john@example.com
```

The form data is sent in the request body rather than being appended to the URL.

Why POST is suitable for registration?

Registration forms usually contain information such as:

- Name
- Email address
- Phone number

- Password
- Address
- Other personal information

Using POST is more appropriate because the submitted data is not directly displayed in the URL.

For example, with GET:

```
/register?name=John&email=john@example.com
```

With POST:

```
/register
```

The data is placed inside the request body.

Comparison with GET

GET	POST
Data is added to URL	Data is sent in request body
Data can be visible in browser history	Data is not displayed in URL
Suitable for retrieving information	Suitable for submitting data
Limited URL length	Supports larger data
Not suitable for passwords	More suitable for sensitive form data

Important security point

Although POST is more appropriate than GET, **POST alone does not encrypt data**.

For secure transmission, the application must use:

```
HTTPS
```

Therefore, the correct combination is:

```
POST + HTTPS
```

Conclusion

POST is the appropriate method for registration because it is designed for submitting data to the server and does not expose the submitted data directly in the URL. For complete security, it must be combined with HTTPS and proper server-side security practices.

3. Choose the proper semantic element for navigation and justify its usage.

Correct Answer: B. `<nav>`

The `<nav>` element is specifically designed to contain major navigation links.

Example:

```
<nav>
    <a href="index.html">Home</a>
    <a href="about.html">About</a>
    <a href="services.html">Services</a>
    <a href="contact.html">Contact</a>
</nav>
```

A navigation menu allows users to move between different pages or sections of a website.

Why `<nav>` is the correct element?

The `<nav>` element provides semantic meaning to the webpage. It tells the browser, search engines, and assistive technologies that the content contains navigation links.

For example, a screen reader can identify:

```
Main Navigation
```

and allow users to quickly navigate through the available links.

Analysis of other options

Option A: `<div>`

```
<div>
    <a href="home.html">Home</a>
</div>
```

A `<div>` is a generic container. It does not indicate that its content represents navigation.

Option C: `<section>`

```
<section>
```

A <section> is used to group related content. It is not specifically intended for navigation.

Option D: **<header>**

```
<header>
```

A <header> is used for introductory content, logos, headings, or other header information. It may contain navigation, but it is not the specific element for navigation.

Example of proper usage

```
<header>
    <h1>My Web Application</h1>

    <nav aria-label="Main Navigation">
        <a href="/">Home</a>
        <a href="/about">About</a>
        <a href="/contact">Contact</a>
    </nav>
</header>
```

Here, <header> defines the introductory area, while <nav> specifically defines the navigation menu.

Conclusion

The <nav> element is the correct semantic element because it clearly identifies navigation content, improves accessibility, supports screen readers, and follows HTML5 standards.

4. Identify the correct XML syntax and explain the error in other options.

Correct Answer: B

```
<note>
    <to>User</to>
    <from>Admin</from>
</note>
```

This is the correct XML structure.

XML requires strict syntax rules. Unlike HTML, XML does not automatically correct errors. Every element must be properly opened and closed, and elements must be correctly nested.

The above structure contains:

```
Root Element: <note>
Child Element: <to>
Child Element: <from>
```

The document has one root element, `<note>`, and all elements are properly closed.

Analysis of Option A

```
<note><to>User</to><from>Admin</from>
```

Error

The root element `<note>` is not closed.

The correct version is:

```
<note>
    <to>User</to>
    <from>Admin</from>
</note>
```

XML requires every non-empty element to have a closing tag.

Analysis of Option C

```
<note><to>User</from></note>
```

Error

The opening element is:

```
<to>
```

but it is closed using:

```
</from>
```

The opening and closing tags must match.

Incorrect:

```
<to>User</from>
```

Correct:

```
<to>User</to>
```

Analysis of Option D

```
<note to="User"></note
```

Error

The closing tag is incomplete because it is missing the > symbol.

Incorrect:

```
</note
```

Correct:

```
</note>
```

The correct XML is:

```
<note to="User"></note>
```

XML well-formedness rules

A valid XML document must follow these rules:

1. It must have exactly one root element.
2. Every opening tag must have a matching closing tag.
3. Elements must be properly nested.
4. XML tags are case-sensitive.
5. Attribute values must be enclosed in quotation marks.
6. Special characters must be properly escaped.

For example:

```
<user name="John">
    <email>john@example.com</email>
</user>
```

Conclusion

Option B is correct because all elements are properly opened, closed, and nested. The other options violate XML well-formedness rules.

5. Choose the correct protocol for web communication and justify your choice.

Correct Answer: B. HTTP

HTTP stands for:

```
HyperText Transfer Protocol
```

It is the primary protocol used for communication between web browsers and web servers.

The communication process is:

```
Web Browser
     |
     | HTTP Request
     ↓
Web Server
     |
     | HTTP Response
     ↓
Web Browser
```

For example, when a user enters:

```
https://example.com
```

the browser sends an HTTP request to the server.

Example:

```
GET / HTTP/1.1
Host: example.com
```

The server then sends a response:

```
HTTP/1.1 200 OK
Content-Type: text/html
```

The response may contain:

- HTML

- CSS
- JavaScript
- Images
- Videos
- Other web resources

Analysis of the other protocols

A. FTP

FTP stands for **File Transfer Protocol**.

It is mainly used to transfer files between computers and servers. It is not the primary protocol used for normal webpage communication.

C. SMTP

SMTP stands for **Simple Mail Transfer Protocol**.

It is used to send email messages between mail servers.

D. SNMP

SNMP stands for **Simple Network Management Protocol**.

It is used to monitor and manage network devices such as:

- Routers
- Switches
- Servers

It is not used for normal browser-to-web-server communication.

HTTP and HTTPS

Modern websites generally use HTTPS:

```
HTTPS = HTTP + Encryption
```

HTTPS provides secure communication by encrypting data exchanged between the browser and server.

This is especially important for:

- Login credentials
- Registration data
- Payment information
- Personal information

Conclusion

HTTP is the correct protocol for web communication because it is specifically designed for communication between web browsers and web servers. HTTPS is preferred for modern applications because it provides encryption and protects data during transmission.

Overall Case Study Conclusion

The startup can solve its web development challenges by correctly applying established web standards.

The correct solutions are:

1. **Video Embedding:** Use the HTML5 `<video>` element.
2. **Registration Form:** Use the `POST` method, preferably over HTTPS.
3. **Navigation:** Use the semantic `<nav>` element.
4. **XML Configuration:** Follow strict XML rules for proper nesting and closing tags.
5. **Web Communication:** Use HTTP, preferably HTTPS for secure communication.

Together, these technologies provide the foundation for a modern web application. HTML5 provides meaningful webpage structure and multimedia support, HTTP enables communication between browsers and servers, XML provides structured configuration data, and HTTPS protects information during transmission. By following these standards, the startup can build a web application that is more secure, accessible, maintainable, and compatible with modern web browsers.